ECE 501: Digital Image Processing

Course Project – Week 5 Progress Report

Adaptive Fabric Surface Defect Detection and Localization using Classical Texture Descriptors

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Project 17, Group 7

Project Title

Adaptive Fabric Surface Defect Detection and Localization using Classical Texture Descriptors

Team Details

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1 Work Done This Week

1.1 Full-Data Feature Set

Last week's results came from a three-image worked example, which was too small to interpret. This week we rebuilt the patch and feature set on the full data: 40 defect-free images and all 103 mask-matched defective images (143 images), with 64×64 patches at stride 32 and the same 68-dimensional descriptor (GLCM 16, LBP 28, Gabor 16, FFT 8). This gives 127,057 patches, of which 677 are defective (prevalence 0.53%, about 1:188). Features are now cached to disk so that a notebook restart no longer loses them. Gabor responses are computed once per image and pooled inside each patch window, which avoids wrap-around artifacts at patch borders.

1.2 Diagnosis of the Weak Baseline

We tested why the baseline scores (mean IoU 0.083, precision 0.094) are low. The main finding is that the evaluation design limits the achievable scores, independently of the descriptors:

- An *oracle* patch detector that flags exactly the patches overlapping the ground truth by at least 10% reaches a mean IoU of only 0.030 at 64×64 patches with stride 64 (median 0). Even 16×16 patches with stride 8 reach only 0.129 (Table 1, Fig. 1). The defects are thin, so for at least half of the defective images no 64×64 patch is labeled defective at all. The 0.083 mean on the three worked-example images is therefore not representative of the full set.
- A 95th-percentile threshold flags about 5% of normal patches by construction, which caps patch-level precision at 0.097 even with perfect recall (Table 2).

- The threshold 9.547 is almost exactly the 95th percentile of a chi distribution with 68 degrees of freedom (9.39), i.e. it describes the training distribution and carries no information about defects. This is consistent with image 0087_030_02 (almost everything flagged) and image 0099_019_02 (missed).

1.3 Descriptor Selection

We scored every single descriptor, every pair and the full set with image-level 5-fold cross-validation repeated over three seeds (15 folds in total; no patches of one image appear in both training and test folds). Each configuration is fitted on defect-free patches with a Ledoit–Wolf-conditioned Mahalanobis score and ranked by average precision (AP; chance level equals the prevalence, 0.0053) and ROC-AUC. The selected pair is **Gabor + FFT** (24 features instead of 68). Section 2 gives the numbers and the evidence for and against this choice.

2 Results

2.1 Evaluation Ceilings

Table 1: Oracle IoU: best possible IoU of a perfect patch classifier (103 mask-matched images)

Patch / stride (px)	Mean IoU	Median IoU
64 / 64	0.030	0.00
64 / 32	0.024	0.00
32 / 16	0.066	0.00
16 / 8	0.129	0.12

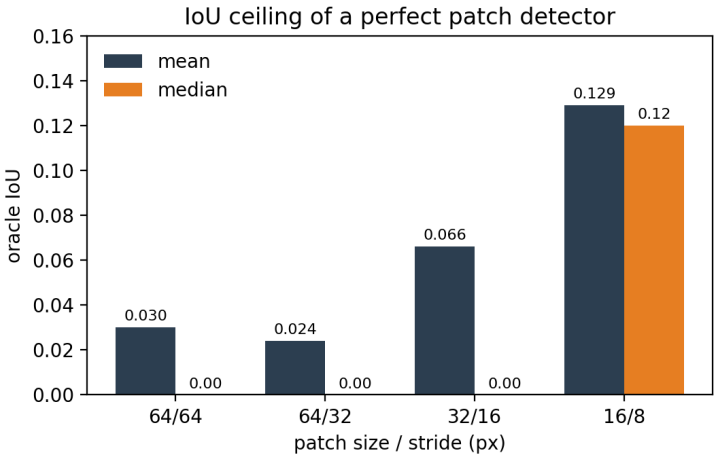


Figure 1: Mean and median IoU achievable by a perfect patch detector at different patch sizes and strides.

Table 2: Maximum patch-level precision at prevalence 0.53% for a given recall (TPR) and false-positive rate (FPR)

	FPR = 5%	FPR = 1%
TPR = 1.00	0.097	0.349
TPR = 0.56	0.057	0.231

2.2 Descriptor Ranking

Table 3: Cross-validated patch-level performance, pooled normal model (mean over 15 folds; chance AP = 0.0053)

Descriptors	AP	AP std	AUC	AP / chance
Gabor + FFT	0.102	0.156	0.906	19.1
GLCM + Gabor	0.087	0.126	0.883	16.3
All minus LBP	0.086	0.134	0.900	16.2
All minus GLCM	0.078	0.126	0.884	14.6
LBP + Gabor	0.077	0.121	0.855	14.5
Gabor alone	0.076	0.100	0.846	14.2
All four (68 features)	0.074	0.113	0.896	13.8
All minus FFT	0.071	0.104	0.891	13.3
All minus Gabor	0.021	0.026	0.824	4.0
GLCM + LBP	0.021	0.024	0.808	4.0
GLCM + FFT	0.021	0.026	0.807	3.9
GLCM alone	0.020	0.021	0.724	3.8
FFT alone	0.013	0.017	0.740	2.5
LBP + FFT	0.013	0.016	0.750	2.4
LBP alone	0.010	0.012	0.695	1.8

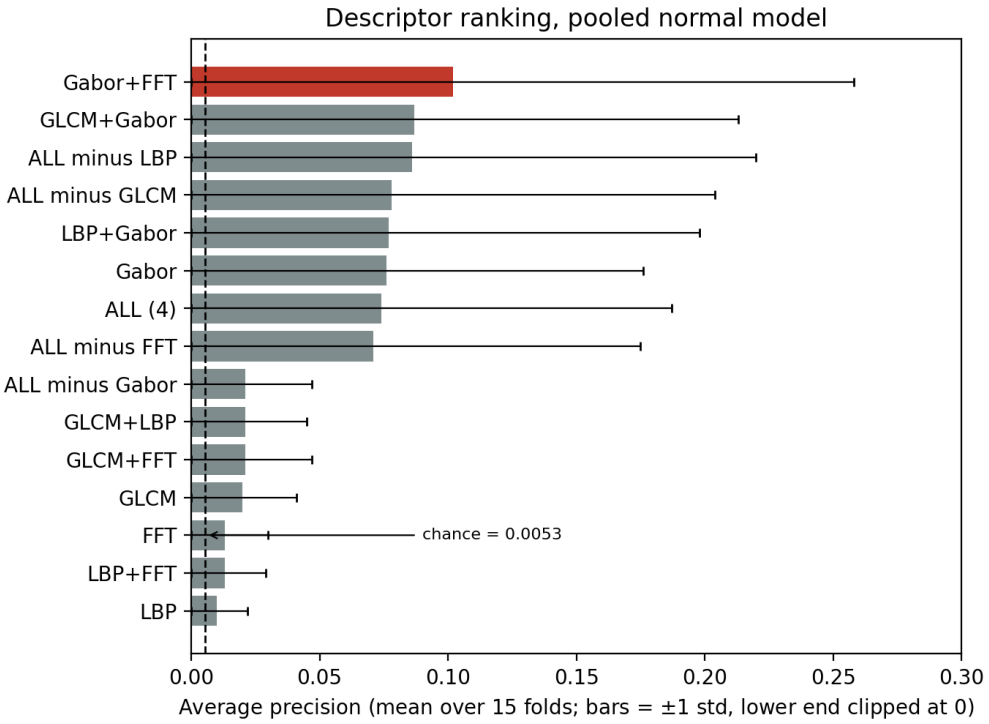


Figure 2: Average precision of every descriptor configuration. The configurations containing Gabor form a clear top group; differences inside that group are small relative to the fold-to-fold spread.

2.3 Validation of the Selected Pair

Table 4: Checks on the selected pair (Gabor + FFT)

Check	Result
Label-permutation test	AP 0.0054 with shuffled labels vs 0.102 real (collapses to chance)
Same pair across CV seeds	Selected in all three seeds
Pair vs Gabor alone	ΔAP +0.026, pair wins in only 47% of folds (not clearly better)
Pair vs FFT alone	ΔAP +0.088, pair wins in 93% of folds
Pair vs all four	ΔAP +0.028, wins in 73% of folds; bootstrap 95% CI $[0.000, +0.055]$
Pair vs GLCM + Gabor	Bootstrap median ΔAP +0.004, 95% CI $[-0.008, +0.023]$ (indistinguishable)
Score correlation (Gabor, FFT)	Spearman 0.24, same as the average over all descriptor pairs (0.235)
Removing a descriptor from the full set	Gabor: AP falls 0.074 $\rightarrow$ 0.021; FFT: -0.003; removing LBP or GLCM does not lower AP (0.086, 0.078)

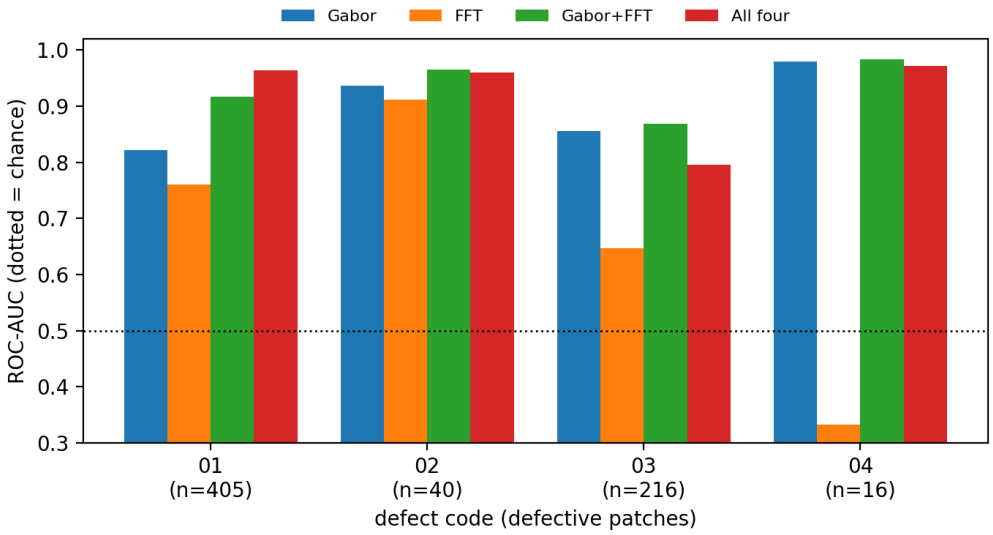


Figure 3: Patch-level ROC-AUC by defect code. Codes with fewer than five defective patches are omitted.

Table 5: Pooled normal model vs. per-image self-referenced normalization (AP / AUC)

Descriptors	Pooled	Self-referenced
Gabor + FFT	0.102 / 0.906	0.009 / 0.694
GLCM + Gabor	0.087 / 0.883	0.010 / 0.734
Gabor alone	0.076 / 0.846	0.011 / 0.711
All four	0.074 / 0.896	0.009 / 0.724
GLCM alone	0.020 / 0.724	0.007 / 0.709
FFT alone	0.013 / 0.740	0.004 / 0.523
LBP alone	0.010 / 0.695	0.006 / 0.590

2.4 Interpretation

- **The anomaly scores carry a real defect signal.** The best configuration reaches patch-level AUC 0.906 and AP 19 times above chance, and the permutation test collapses to chance. The weak IoU and precision of the baseline are therefore largely a consequence of the evaluation design (Tables 1 and 2).

- **Gabor is the strongest descriptor.** Every top-ranked configuration contains it, and removing it from the full set removes most of the performance (Table 3, Fig. 2).

- **FFT is a weak second choice.** Gabor + FFT has the highest AP, and FFT is the cheapest partner (8 features) and encodes the global periodic structure of the weave. However, the pair does not beat Gabor alone in most folds and is not statistically separable from GLCM + Gabor, so we report it as the best-scoring partner, not a proven better one. Fold-to-fold variation is large (AP std 0.156 against a mean of 0.102).

- **Defect types behave differently.** The pair has the highest AUC on codes 02, 03 and 04, but the full set is better on code 01, the largest class (AUC 0.964 vs 0.917), and worse on code 03 (0.796 vs 0.869) (Fig. 3). Codes 02 and 04 have only 40 and 16 defective patches, so those values are uncertain.

- **Per-image normalization did not help.** All configurations drop to AP of about 0.01 and

AUC of about 0.7 (Table 5), so the pooled model is kept for now.

3 Work To Be Done

- Correct the fabric-ID parser against the AFID naming convention (parsed IDs exceed the seven AFID fabric types) and re-run the worked example with per-fabric models (first priority).
- Refit the anomaly model on the selected Gabor + FFT features (24 dimensions) and re-run localization on all 103 mask-matched images.
- Replace pixel-level IoU as the headline measure with patch-level AP and ROC-AUC and a defect-level measure (a defect counts as found if any flagged patch overlaps it, plus false alarms per image).
- Relax the 10% overlap labeling rule (e.g. label a patch defective on any overlap) and test smaller patches.
- Compare the Ledoit–Wolf estimator directly against the original ridge-regularized covariance.
- Fit and compare per-fabric thresholds against the single pooled threshold.
- Add morphological post-processing (opening/closing) to reduce over-triggering such as in image 0087_030_02.
- Check the grayscale assumption explicitly, since some sampled images reported up to four channels.
- Map numeric defect codes to AITEX's named defect categories.

4 Literature Review

No change since the mid-semester report: the AFID dataset paper [1], the GLCM [2] and LBP [3] texture-descriptor papers, and a Gabor-filter texture-feature reference [4]. A texture-feature-on-AITEX benchmarking reference remains a candidate addition before end-semester.

5 Challenges Faced

- A notebook restart lost the in-memory feature matrix, so we rebuilt the patch set from the raw data and now cache all features to disk.
- Much of the weak baseline is structural: the oracle IoU is 0.03 at 64×64 patches with stride 64 and the 5% false-positive rate caps precision near 0.1, so the evaluation, not only the model, needs to change.
- The descriptor selection is only partly conclusive: Gabor is clearly the strongest descriptor, but the second descriptor (FFT) is not statistically separable from GLCM, results vary strongly between folds, and only four defect codes had enough defective patches for a per-type analysis.
- The fabric-ID parsing issue means per-fabric modeling has not yet been properly tested; fixing it is a prerequisite for the next experiments.

References

[1] J. Silvestre-Blanes, T. Albero-Albero, I. Miralles, R. Pérez-Llorens, and J. Moreno, "A public fabric database for defect detection methods and results," *Autex Research Journal*, vol. 19, no. 4, pp. 363–374, 2019.

[2] R. M. Haralick, K. Shanmugam, and I. Dinstein, "Textural features for image classification," *IEEE Trans. Syst., Man, Cybern.*, vol. SMC-3, no. 6, pp. 610–621, 1973.

[3] T. Ojala, M. Pietikäinen, and T. Mäenpää, "Multiresolution gray-scale and rotation invariant texture classification with local binary patterns," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 24, no. 7, pp. 971–987, 2002.

[4] B. S. Manjunath and W. Y. Ma, "Texture features for browsing and retrieval of image data," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 18, no. 8, pp. 837–842, Aug. 1996.